

AMENDMENT NO. 2 JUNE 2024

TO

IS 8034 : 2018 SUBMERSIBLE PUMPSETS — SPECIFICATION

(*Third Revision*)

(*Second cover ,foreword*) — Delete (e) and add following before the last paragraph:

‘Graphs for calculating minimum pump efficiency are replaced by two separate equations. Equation (1) is for pump efficiency of 100 mm borewell pumps and 150 mm borewell pumps, 2 pole speed and equation (2) is for pump efficiency of 200 mm and above borewell pumps, 2 pole speed’

[*Page 8, Table 1, Notes, SI No. (6)*] — Substitute the following for the existing:

‘6 Motor efficiency factors shall be applied to the pump efficiency calculated as per 11.4.1(a) to arrive at the minimum-overall efficiency of the pumpset.’

[*Page 8, Table 2, Notes, SI No. (6)*] — Substitute the following for the existing:

‘6 Motor efficiency factors shall be applied to the pump efficiency calculated as per 11.4.1(a) to arrive at the minimum-overall efficiency of the pumpset.’

[*Page 9, Table 3, Notes, SI No. (6)*] — Substitute the following for the existing:

‘6 Motor efficiency factors shall be applied to the pump efficiency calculated as per 11.4.1(b) to arrive at the minimum-overall efficiency of the pumpset.’

[*Page 10, Table 4, Notes, SI No. (4)*] — Substitute the following for the existing:

‘4 Motor efficiency factors shall be applied to the pump efficiency calculated as per 11.4.1(a) to arrive at the minimum-overall efficiency of the pumpset.’

[*Page 10, Table 5, Notes, SI No. (4)*] — Substitute the following for the existing:

‘4 Motor efficiency factors shall be applied to the pump efficiency calculated as per 11.4.1(a) to arrive at the minimum-overall efficiency of the pumpset.’

(*Page 10, Table 5*) — Insert the following new table at the end:

Table 6 Values of Performance Characteristics for 2-Pole, 415 Volt, 50 Hz Three Phase, Oil Filled Submersible Motors for Bore Size Nominal 150 mm Shall Have Maximum OD of Motor 146 mm

(*Clauses 11.3.1 and 11.4.3*)

Sl No.	Rated Output kW	Minimum Full Load Speed rpm	Maximum Full Load Current A	Permissible Limit of Maximum Current in the Operating Head Range for Checking the Non-overloading Requirements A	Motor Efficiency Factor
(1)	(2)	(3)	(4)	(5)	(6)
i)	2.2	2 810	5.8	6.21	72
ii)	3.0	2 810	7.8	8.35	73
iii)	3.7	2 830	9.1	9.74	74
iv)	4.5	2 830	11.1	11.88	76
v)	5.5	2 850	13.8	14.77	79
vi)	7.5	2 850	18.0	19.26	81

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Sl No.	Rated Output kW	Minimum Full Load Speed rpm	Maximum Full Load Current A	Permissible Limit of Maximum Current in the Operating Head Range for Checking the Non-overloading Requirements A	Motor Efficiency Factor
(1)	(2)	(3)	(4)	(5)	(6)
vii)	9.3	2 870	23.4	25.04	81
viii)	11.0	2 870	27.3	29.21	82
ix)	13.0	2 870	32.0	34.24	83
x)	15.0	2 880	36.7	39.27	84

NOTES

- 1 Maximum current limits specified are for 415 V rated voltage. For other voltages, it shall be in inverse proportion to rated voltage.
- 2 For three phase motors, the value of current shall be taken as average value of the current measured in three phases.
- 3 Values given in col (5) are 1.07 times of the values given in col (4).
- 4 Performance values of 4-pole three phase motors and 2-pole three phase motors of less than 1.1 kW and exceeding 15.0 kW rating shall be as declared by the manufacturer.
- 5 However, for three phase motors above 15.0 kW rating, the declared value of efficiency shall not be lesser than that of 15.0 kW efficiency value.
- 6 Motor efficiency factors shall be applied to the pump efficiency calculated as per **11.4.1(a)** to arrive at the minimum overall efficiency of the pumpset.

(Page 11, clause **10.2**) — Substitute ‘Fig. 3’ for ‘Fig. 4’

(Page 11, clause **11.1.2**, line 4) — Substitute ‘However, after applying the tolerance, overall efficiency value shall not be less than obtained after multiplying the motor efficiency factor and pump efficiency as derived from equation (1) and equation (2) of **11.4**.’ for ‘However, after applying the tolerance, overall efficiency value shall not be less than that calculated from Fig. 4 and Fig. 5.’

(Page 13, Fig. 4) — Delete.

(Page 14, Fig. 5) — Delete.

(Page 14, clause **11.1.5**, line 4 and 10) — Substitute ‘col (4) of Tables 1 to 5 and col (5) of Table 6’ for ‘col (4) of Tables 1 to 5’

(Page 15, clause **11.3.1**, line 2) — Substitute ‘Tables 1, 2, 3, 4, 5 and 6’ for ‘Tables 1, 2, 3, 4, and 5’

(Page 15, clause **11.4**) — Substitute the following for the existing:

‘11.4 Pump Efficiency

11.4.1 Pump efficiency shall be calculated using the equation as given below, and efficiency corresponding to MEL 0.2 shall be taken as the minimum pump efficiency:

- a) Pump efficiency for 100 mm borewell pumps and 150 mm borewell pumps, 2 pole speed:

$$\eta_p = 0.5118 + 0.841y - 0.00192x^2 + 0.01095y^2 - 0.01841xy + C \dots (1)$$

where

- η_p = minimum pump efficiency;
 Y = q ;
 X = n_s ;
 n_s = specific speed;
 C = values for different pump efficiency levels.
 Q = discharge rate in m³/h; and

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Minimum efficiency level, MEL	0.20	0.30	0.40	0.50	0.60	0.70
C	38	40	42	44.1	46.3	48.6

NOTES

- 1 Other MEL values are for the guidance of the manufacturer and to upgrade to higher values of pump efficiency.
- 2 An illustrative example for calculating pump efficiency using above equation is given in Annex B.
- 3 The calculated value for specific speed and minimum pump efficiency shall be rounded off up to 2 decimal places.

b) Pump efficiency for 200 mm and above borewell pumps, 2 pole speed:

$$\eta_p = 0.657 \ 1 \ x + 0.085 \ 1 \ y - 0.005 \ 34 \ x^2 - 0.000 \ 565 \ y^2 + 0.001 \ 06 \ xy + C \dots\dots\dots (2)$$

where

- η_p = minimum pump efficiency;
 Y = q ;
 X = n_s ;
 C = values for different pump efficiency levels.
 n_s = specific speed;
 Q = discharge rate in m³/h; and

Minimum efficiency level, MEL	0.20	0.30	0.40	0.50	0.60	0.70
C	42.0	44.0	45.9	47.7	49.4	51.0

NOTES

- 1 Other MEL values are for the guidance of the manufacturer and to upgrade to higher values of pump efficiency.
- 2 An illustrative example for calculating pump efficiency using above equation is given in Annex C.
- 3 The calculated value for specific speed and minimum pump efficiency shall be rounded off up to 2 decimal places.

$$n_s = n \cdot \frac{\sqrt{Q/3 \ 600}}{(H)^{0.75}}$$

where

- n = rotational speed, in revolutions per minute (rpm);
 Q = discharge rate, in m³/h; and
 H = Head, per stage in meters (m).

11.4.2 The efficiency calculated as per **11.4.1(a)** and **11.4.1(b)** represents three or more stages.

- a) For two stage pump, multiply efficiency calculated by a factor 0.98; and
- b) For single stage pump, multiply efficiency calculated by a factor 0.97.

11.4.3 The motor efficiency factor of motor ratings not given in the Tables 1, 2, 3, 4, 5 and 6 shall be declared by the manufacturer. The motor efficiency factor of motors used with pumpsets suitable for bore sizes more than 200 mm shall be as declared by the manufacturer, but it shall be not less than the motor efficiency factor of motors of same rating for 200 mm bore size.

11.4.4 For overall efficiency of the pumpset, multiply pump efficiency calculated as per **11.4.1(a)** or **11.4.1(b)** by corresponding motor efficiency factor as given in Tables 1, 2, 3, 4, 5 and 6.

11.4.5 The efficiency equations include non-return valve losses.'

(Page 15, clause **11.5.1**) — Substitute the following for the existing:

'**11.5.1** For arriving at minimum overall efficiency of the pumpset, multiply minimum pump efficiency calculated as per **11.4.1(a)** or **11.4.1(b)** by corresponding motor efficiency factor as per Table 1 to Table 6.'

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(Page 18, Annex A) — Insert the following at the end:

‘ANNEX B

(Clause 11.4.1)

EXAMPLE OF MINIMUM PUMP EFFICIENCY CALCULATION FOR 150 MM BOREWELL PUMPS

B-1 INPUT DATA

Head = 35 m
Discharge = 6.5 lps
Speed = 2 800 rpm
Stages = 05

B-2 CALCULATIONS

Head per stage = $35/5 = 7$ m.

B-2.1 To Find out Discharge Rate, Q, in m³/h

Discharge = $6.5 \text{ lps} \times 3.6 = 23.4 \text{ m}^3/\text{h}$

$$n_s = n \cdot \frac{\sqrt{Q/3600}}{(H)^{0.75}}$$

$$n_s = 2800 \times \frac{\sqrt{23.4/3600}}{(7)^{0.75}} = 52.45$$

NOTE — For specific speed calculation we have to use discharge (Q) in m³/h and head per stage (H) in m.

$$\eta_P = 0.5118x + 0.841y - 0.00192x^2 + 0.01095y^2 - 0.01841xy + C \dots(1)$$

$$x = n_s = 52.45$$

$$y = Q = 23.40$$

NOTE — For minimum pump efficiency calculation we have to use discharge (Q) in m³/h.

C = 38 (C-value for MEL 0.2 to be taken from the table for minimum pump efficiency calculation)

$$\eta_P = 0.5118 \times 52.45 + 0.841 \times 23.4 - 0.00192 \times 52.45^2 + 0.01095 \times 23.4^2 - 0.01841 \times 52.45 \times 23.4 + 38$$

$$\eta_P = 62.60 \text{ percent (minimum pump efficiency)}$$

ANNEX C

(Clause 11.4.1)

EXAMPLE OF MINIMUM PUMP EFFICIENCY CALCULATION FOR 200 MM AND ABOVE BOREWELL PUMPS, IS 8034

C-1 INPUT DATA

Head = 32 m
Discharge = 6.5 lps
Speed = 2 900
Stages = 02

C-2 CALCULATIONS

Head per Stage = $32/2 = 16$ m

C-2.1 To Find out Discharge Rate, Q, in m³/h

Discharge = $6.5 \text{ lps} \times 3.6 = 23.4 \text{ m}^3/\text{h}$

$$n_s = n \cdot \frac{\sqrt{Q/3600}}{(H)^{0.75}}$$

$$n_s = 2900 \times \frac{\sqrt{23.4/3600}}{(16)^{0.75}} = 29.22$$

NOTE — For specific speed calculation we have to use discharge (Q) in m³/h and head per stage (H) in m.

$$\eta_P = 0.6571x + 0.0851y - 0.00534x^2 - 0.000565y^2 + 0.00106xy + C \dots\dots(2)$$

$$x = n_s = 29.22$$

$$y = Q = 23.40$$

NOTE — For minimum pump efficiency calculation we have to use discharge (Q) in m³/h.

C = 42 (C-value for MEL 0.2 to be taken from the table for minimum pump efficiency calculation)

$$\eta_P = 0.6571 \times 29.22 + 0.0851 \times 23.4 - 0.00534 \times 29.22^2 - 0.000565 \times 23.4^2 + 0.00106 \times 29.22 \times 23.4 + 42$$

$$\eta_P = 59.05 \text{ percent} \times 0.98 \text{ (factor for 2 stage)}$$

$$\eta_P = 57.86 \text{ percent (Minimum pump efficiency)'}.$$